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A compact and inducible dCas12f-based CRISPRa platform for programmable in vivo gene activation

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Precise activation of endogenous genes is a powerful strategy for functional genomics and therapeutic development, but current CRISPR-based transcriptional activation (CRISPRa) systems are limited by the large size of Cas proteins for adeno-associated virus (AAV) delivery. Here, we present a high-efficiency dCas12f-based transcriptional activation system (HEAL), which recruits transactivators through MS2 coat protein binding to MS2 aptamers embedded within the sgRNA scaffold. Engineered to enhance DNA binding, nuclear localization, and transactivator recruitment, HEAL induces over 100,000-fold activation of endogenous genes and outperforms existing CRISPRa systems in vitro and in vivo. We further develop red-light-inducible OptoHEAL and small-molecule-inducible ChemHEAL for remote and precise transcriptional control. AAV-delivered HEAL targeting *interleukin 10* alleviates acute kidney injury in mice, while ChemHEAL-mediated activation of *thymic stromal lymphopoietin* reduces body weight in obese mice. HEAL provides a modular, compact, and controllable platform for endogenous gene activation with strong potential for fundamental research and gene therapy.

Transcriptional regulation orchestrates complex gene expression programs that govern development, maintain cellular homeostasis, and mediate adaptive responses to environmental cues[1,2]. Dysregulation of these networks drives a causal factor in a multitude of pathological states, including neurodevelopmental disorders[3,4], immune dysregulation[5], and metabolic syndromes[6]. CRISPR-based transcriptional activation (CRISPRa) systems, which repurpose catalytically inactive Cas (dCas) proteins to recruit transactivators to specific genomic loci, have been widely developed to enable targeted modulation of endogenous gene expression for functional studies and therapeutic exploration[7–12].

While dCas9- and dCas12a-based CRISPRa systems have successfully demonstrated gene activation, their large coding size (~4.0–5.5 kb) poses a substantial obstacle for in vivo delivery, particularly via adeno-associated virus (AAV) vectors with a ~4.7 kb packaging limit[13,14]. Efforts to address this limitation have driven the development of compact CRISPR systems based on miniature Cas proteins such as Cas12f[15–20]. The dCasMINI system, based on Cas12f, has demonstrated in vivo gene activation via AAV delivery[21,22]. Yet current Cas12f-based tools remain suboptimal in activation strength and lack dynamic control or modular programmability, limiting their use in contexts requiring precise, tunable gene regulation. A compact and programmable CRISPRa platform with efficient and controllable activation remains a critical need for both basic and therapeutic applications.

The safety and effectiveness of therapeutic applications of CRISPRa systems demand precise regulation of gene expression dynamics, particularly in complex physiological and pathological contexts[23,24].

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Sustained or uncontrolled in vivo gene activation can lead to adverse effects, exemplified by the tight physiological regulation of genes such as *OPA1* (*optic atrophy 1*), *RAI1* (*retinoic acid-induced 1*), *MeCP2* (*methyl-CpG-binding protein 2*), and *TSLP* (*thymic stromal lymphopoietin*), the dysregulation of which has been linked to toxicity or developmental abnormalities[25–29]. Accordingly, inducible CRISPRa systems based on dCas9 or dCas12a responsive to exogenous cues-including small molecules[30,31], light[32–36], or ultrasound[37]-have been developed for dynamic and reversible control of gene expression. Such strategies enable quantitative modulation of target gene expression at predefined levels, offering promise for dosage-sensitive therapeutic applications. However, their reliance on large Cas proteins poses a challenge for AAV packaging, consequently limiting their efficiency and applicability in vivo[38].

In this study, we develop a compact and precisely controllable platform for endogenous gene activation. We first engineer a high-efficiency CRISPRa system comprising an engineered ΔdCas12f protein guided by a restructured sgRNA for targeting endogenous loci and transactivators fused to an MS2 coat protein (MCP) that are recruited by MS2 aptamers embedded within the sgRNA to activate target gene transcription. The resulting high-efficiency dCas12f-based transcriptional activation (HEAL) system is compatible with single-AAV delivery and demonstrated greater efficiency in activating several endogenous genes than other CRISPRa systems. Further, we develop the red light-responsive OptoHEAL and chemically-inducible ChemHEAL systems, which enable user-defined, finely tunable spatiotemporal regulation of gene activation in vivo. The present study demonstrates the application of these systems in disease-relevant contexts by targeting *Il10* with an AAV-delivered HEAL to effectively alleviate acute kidney injury in mice and activating *Tslp* with an abscisic acid-inducible ChemHEAL to reduce body weight in obese mice. Overall, our findings demonstrate the broad potential of HEAL as a compact, modular, and controllable CRISPRa platform for basic biological studies and therapeutic applications.

Results

Design and construction of a high-efficiency dCas12f-based transcriptional activation system

To establish a potent and compact CRISPRa platform compatible with single AAV delivery, we developed a high-efficiency dCas12f-based transcriptional activation system (HEAL) by redesigning components of the Un1Cas12f1_ge4.1 framework[39]. HEAL incorporates three strategically engineered features: (1) a restructured sgRNA scaffold with improved architecture to promote efficient recruitment of transactivators, (2) a modified dCas12f variant with enhanced DNA binding affinity and nuclear localization, and (3) a modular assembly of potent transactivation domains to amplify endogenous gene activation (Fig. 1a).

Building upon previous findings that RNA aptamer-mediated recruitment of transactivator can outperform direct fusion to dCas proteins[10,40], we redesigned the sgRNA scaffold by incorporating MS2 aptamer structures to enable modular effector recruitment. Through structural analysis, we identified five potential engineering sites (ES1–ES5) within the scaffold (Fig. 1b) and introduced MS2 insertions and base substitutions to enhance transactivator recruitment and scaffold stability. Initial screening in HEK-293T cells revealed that MS2 aptamers inserted at ES1 and ES2 conferred the strongest activation, achieving up to a 1453-fold increase in transcription of the endogenous *hemoglobin subunit gamma (HBG)* gene (Fig. 1c). Further improvement was achieved by a single-base substitution (A → C) at ES5, which increased scaffold stability and enhanced *HBG* activation (Fig. 1d and Supplementary Fig. 1), demonstrating the additive effects of scaffold architecture refinement.

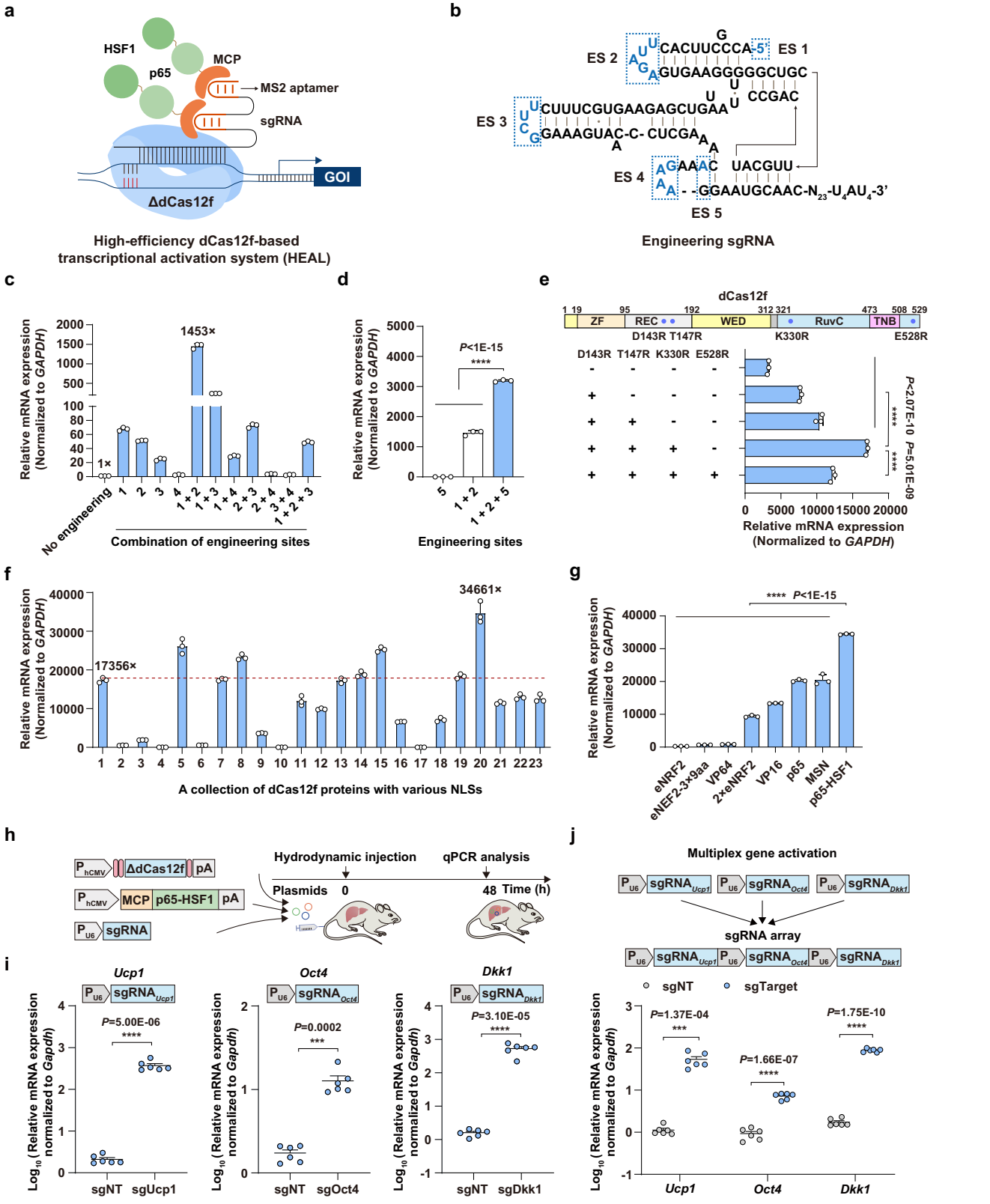
In parallel, we engineered the dCas12f protein to improve both DNA binding affinity and nuclear localization efficiency. Building on structural modeling analyses and insights from the engineering of CasMINI[15], we introduced targeted amino acid substitutions into dCas12f to enhance its DNA-binding capacity. A triple mutant variant (D143R, T147R, and K330R) exhibited the greatest enhancement in transcriptional activation (Fig. 1e). To improve nuclear localization, we evaluated multiple nuclear localization signal (NLS) configurations fused to the mutated dCas12f (Supplementary Fig. 2a). The configuration containing two c-MYC NLSs at the N-terminus and one at the C-terminus—designated ΔdCas12f—resulted in the highest activation efficiency, leading to a 34,661-fold upregulation of *HBG* expression (Fig. 1f). In addition, we screened a panel of transcriptional activation domains (TADs) fused to MCP, including VP16 and its tetrameric variant VP64 from herpes simplex virus type 1 (HSV-1)[41], as well as chimeric activators derived from human mechanosensitive transcription factors, such as eNRF2 and MSN[42] (Supplementary Fig. 2b). Among the candidates, the p65-HSF1 fusion consistently produced the strongest gene activation (Fig. 1g), and was selected for incorporation into the final HEAL system. Collectively, these design efforts led to the establishment of a compact, modular, and highly potent CRISPRa system composed of an engineered dCas12f variant and a restructured sgRNA scaffold. Leveraging HEAL as a platform, we further developed inducible transcriptional activation systems applicable to both in vitro and in vivo settings.

HEAL-mediated endogenous gene activation in mammalian cells and mice

Given the above results showing that the HEAL system could upregulate the endogenous gene, *HBG*, by over 30,000-fold in HEK-293T cells, we next investigated the ability of this system to activate additional endogenous genes across different species and cell lines. We tested six genes in human HEK-293T cells, including *HBG*, *achaete-scute homolog 1 (ASCL1), interferon gamma (IFNγ), rhox homeobox family member 2 (RHOXF2), uncoupling protein 1 (UCP1)*, and *hemoglobin subunit beta (HBB)*, as well as five genes in mouse B16F10 cells, including *Ucp1*, *octamer-binding transcription factor 4 (Oct4), dickkopf-related protein 1 (Dkk1), Tslp*, and *interleukin 10 (Il10)*, and one gene, *follistatin (Fst)*, in mouse N2a cells. For each gene, we designed ten sgRNAs targeting regions upstream and downstream of the transcription start site (TSS), all using the TTTR (TTTA or TTTG) PAM (protospacer adjacent motif) sequence. The degree of activation varied markedly across targets, ranging from >10-fold to nearly 100,000-fold, with *HBG* exhibiting the highest induction (Supplementary Fig. 3a–f and Supplementary Fig. 4a–f). Notably, activation of gene expression was highly dependent on the sgRNA target site (Supplementary Fig. 3a–f and Supplementary Fig. 4a–f). The observed differences may be related to the intrinsic epigenetic state of the gene, chromatin accessibility, and baseline expression levels within the cells[43,44]. We further evaluated the activation potential of HEAL in additional human cell lines, and qPCR analysis showed significant upregulation of *HBG* and *IFNγ* across diverse cellular contexts (Supplementary Fig. 3g, h). We then assessed the ability of the HEAL system to induce the simultaneous activation of multiple endogenous genes. To this end, we constructed an sgRNA array targeting the promoters of *Ucp1*, *Oct4*, and *Dkk1*, and delivered it into B16F10 cells. QPCR analysis showed that the HEAL system enabled activation of multiplexed endogenous genes (Supplementary Fig. 4g). HEAL-mediated gene activation was also validated at the protein level for selected targets, including human IFNγ and murine IL-10 and TSLP (Supplementary Fig. 4h–j). Together, these findings demonstrate the potential of HEAL for user-defined, multiplexed activation of endogenous genes of interest.

To test whether the HEAL system could activate endogenous genes in vivo, we delivered HEAL-encoding plasmids targeting each gene individually by hydrodynamic tail vein injection in C57BL6 mice (Fig. 1h). At 48 h after injection, the mice were euthanized.

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Quantification of *Ucp1*, *Oct4*, and *Dkk1* expression in liver tissues by qPCR showed that the HEAL system significantly upregulated transcription of three respective genes (Fig. 1i). Notably, delivery of HEAL-encoding plasmids with the sgRNA array resulted in successfully activating the expression of all three genes simultaneously in mouse livers (Fig. 1j), highlighting the effectiveness of the HEAL system for endogenous gene activation in both mammalian cells and mice.

Comparisons of the HEAL system with other transcriptional activation systems

We next compared the efficiency of transcriptional activation using the HEAL system with that of other current gene activation systems. To facilitate side-by-side comparison with the dCasMINI and dCpf1 activation systems, we used the same sgRNA seed sequence across all three systems, as they all recognize the TTTR PAM (Fig. 2a). In five gene

Fig. 1 | Construction and characterization of the HEAL system. a Schematic of the HEAL system. CRISPR-ΔdCas12f is assembled with an engineered sgRNA for targeting endogenous or exogenous genomic loci; transactivators fused to MS2 coat protein (MCP) are recruited to the MS2 aptamer embedded within the sgRNA to initiate gene transcription. **b** Schematic of the sgRNA structure illustrating the MS2 aptamer insertion and single-base mutation sites. The MS2 aptamer is either inserted at the engineering site (ES) 1 or replaces the original sequence at ES 2-4. An A-to-C mutation is introduced at ES 5, with a single-base mutation site highlighted in blue. **c**, **d** Screening of sgRNA constructs targeting the endogenous *hemoglobin subunit gamma* (*HBG*) gene. HEK-293T cells were co-transfected with plasmids encoding dCas12f, MCP-p65-HSF1, and sgRNA constructs with different combinations of ES 1-5 (**c**). The sgRNA (ES 1 + 2) was further optimized by introducing an A-to-C mutation at ES 5, and a minimal variant containing only this mutation was also tested (**d**). **e** Screening of dCas12f variants. Mutation sites are indicated in the structural diagram above. **f** Optimization of the dCas12f nuclear localization signal (NLS). The detailed genetic configurations of plasmids encoding dCas12f are provided in Supplementary Fig. 2a. **g** Screening of transcriptional activation domains. In panels (**c**–**g**), qPCR analysis of *HBG* mRNA levels were performed at 48 h post-transfection, and the expression components used are listed in Supplementary Table 2. **h** Schematic of the timeline for HEAL-mediated endogenous gene activation in mice. Plasmids encoding the HEAL system with individual sgRNAs targeting different endogenous genes were delivered to mouse livers via hydrodynamic tail vein injection. **i** Relative mRNA expression of the *Ucp1*, *Oct4*, and *Dkk1* genes in mice. **j** HEAL-mediated simultaneous activation of multiple endogenous genes in mice. An empty sgRNA vector was used as a non-targeted control (sgNT). Data in (**c**–**g**) are means ± s.d.; $n = 3$ independent experiments, with 3 technical replicates for each. Data in (**i**, **j**) are means ± s.e.m.; $n = 6$ mice. Statistical significance was determined by one-way ANOVA (**d**, **e**, **g**), two-tailed unpaired *t* test (**i**, **j**); $^{***}P < 0.001$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

activation tests (two in B16F10 cells and three in HEK-293T cells), HEAL consistently outperformed both dCasMINI and dCpf1, achieving higher maximal activation with fold increases ranging from 3.7- to 537-fold (Fig. 2b–f). We then compared HEAL to a *Streptococcus pyogenes*-derived dCas9-SAM system, which is considered one of the most potent dCas9-based transcriptional activation systems[40] (Fig. 2g). Since dCas9 recognizes the NGG PAM, we designed five sgRNAs for the dCas9-SAM system, targeting different genomic sites for each gene, and two sgRNAs for HEAL. QPCR analysis revealed that HEAL drove markedly stronger gene activation—by 40-fold for *UCP1* and 221-fold for *HBG*. For other targets such as *Ucp1* and *Tslp*, the enhancement was more modest but still evident (2.8- and 7.6-fold, respectively) (Fig. 2h–k).

To assess transcriptional activation performance in vivo, we constructed a reporter that could be targeted by all tested CRISPRa systems at the same locus (Supplementary Fig. 5a), effectively minimizing differences in activation efficiency caused by variations among different target sites. Mice were then hydrodynamically injected with plasmids encoding HEAL, dCasMINI, dCpf1, or dCas9-SAM, along with the luciferase reporter. Bioluminescence imaging at 24 h post-injection revealed that the HEAL system induced significantly stronger reporter expression compared to the other groups, with signal intensities 5.6-, 4.1-, and 2.1-fold higher than those achieved by dCpf1, dCasMINI, and dCas9-SAM, respectively. In contrast, all NT control groups exhibited minimal background activity (Supplementary Fig. 5b–d). In addition, we assessed the in vivo transcriptional activation of the endogenous *Ucp1* gene. In mouse livers harvested 48 h after injection (Fig. 2l), qPCR analysis showed that the HEAL system induced a significantly higher upregulation of *Ucp1* compared to other systems, while maintaining high activation specificity (Fig. 2m). Collectively, these findings highlight the improved efficiency and preserved specificity of HEAL in transcriptional activation over that of existing CRISPRa systems.

HEAL-mediated *Il10* activation for amelioration of acute kidney injury in mice

To investigate the therapeutic potential of the HEAL system, we evaluated its ability to mitigate disease progression in a mouse model of acute kidney injury (AKI) by specifically targeting *interleukin 10* (*Il10*), an anti-inflammatory cytokine known to alleviate cisplatin-induced renal injury[45,46]. As our above results showed that the HEAL system could efficiently activate *Il10* expression in murine cell lines using $sgRNA_{Il10}$-4 (Supplementary Fig. 4e), we constructed a single AAV vector encoding the HEAL system with $sgRNA_{Il10}$-4 (AAV_{Il10}) and administered it to C57BL6 mice via tail vein injection (Fig. 3a).

Fourteen days after AAV injection, AKI was induced through cisplatin treatment and mice were subsequently euthanized four days later. Compared to the untreated AKI group, *Il10* expression in the liver was upregulated by 17.5-fold in the AAV_{Il10} treatment group, accompanied by a 3.5-fold increase in serum IL-10 levels (Fig. 3b, c). Further, *Il10* activation in the AAV_{Il10}-treated AKI mice alleviated cisplatin-induced renal injury, as evidenced by significant reductions in blood urea nitrogen (BUN) levels (from 270.2 ± 46.4 mg/dL to 162.6 ± 32.9 mg/dL) and serum creatinine (S-Cre) levels (from 113.9 ± 25.1 μmol/L to 65.0 ± 18.2 μmol/L), compared to the untreated AKI group (Fig. 3d, e). Immunohistochemical staining revealed that renal injury marker, neutrophil gelatinase-associated lipocalin (NGAL), was significantly reduced in the AAV_{Il10}-treated mice relative to the control groups (Fig. 3f). Moreover, TUNEL staining of kidney sections revealed a lower proportion of apoptotic cells in the AAV_{Il10}-treated group, indicating reduced cellular damage (Fig. 3g, h). Collectively, these findings suggested that HEAL-mediated *Il10* activation could effectively alleviate pathological symptoms in AKI mice, underscoring its potential as a therapeutic strategy for ameliorating kidney damage.

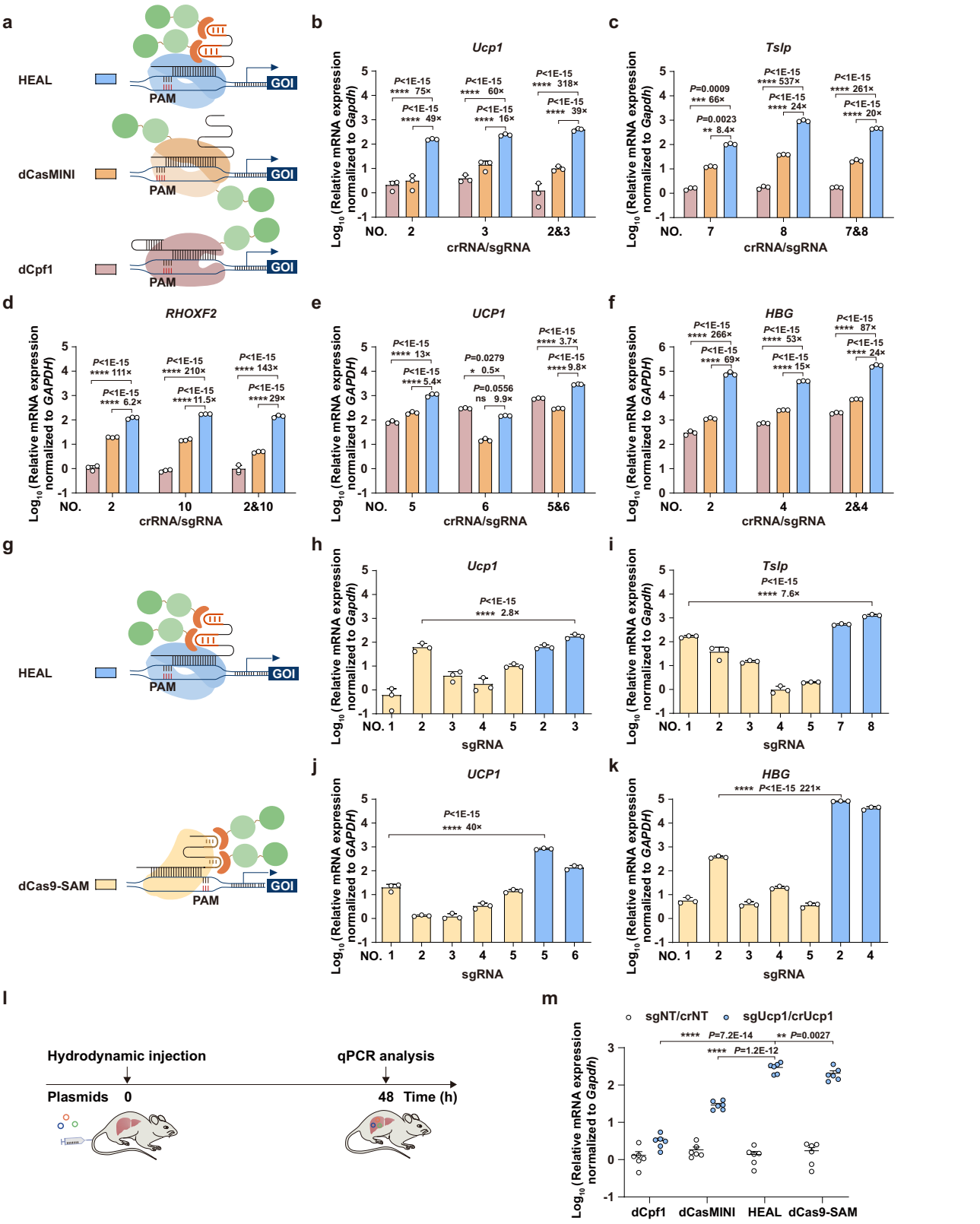
Red light-inducible endogenous gene activation in mice

Optogenetic technologies enable remote, traceless, and reversible control of endogenous gene expression[47], we next sought to apply the HEAL system in a light-inducible, customizable endogenous gene activation system by integrating our previously designed REDLIP system[48]. The resulting red light-inducible HEAL system, OptoHEAL, employs a dimerization-based assembly mechanism for the MCP and transactivator p65-HSF1, wherein exposure to red light (660 nm) induces *Dr*BphP and LDB3 dimerization, facilitating p65-HSF1 recruitment to the target site to initiate gene transcription. Conversely, far-red light (780 nm) disrupts *Dr*BphP-LDB3 interactions, leading to dissociation of p65-HSF1 and termination of transcription (Fig. 4a).

Testing eight different combinations for fusion of the transcriptional activation modules (MCP and p65-HSF1) with light-responsive dimerization modules (*Dr*BphP and LDB3) indicated that the MCP-*Dr*BphP and LDB3-p65-HSF1 combination resulted in the greatest increase (12-fold) in endogenous *HBG* expression between light and dark conditions (Fig. 4b, c). To further improve induction efficiency, we introduced nuclear localization signals (NLS) derived from the SV40 large T antigen. Increasing NLS copy number at the N-terminus of LDB3-p65-HSF1 significantly enhanced light-induced *HBG* mRNA expression, which reached 50-fold higher expression compared to the dark state (Supplementary Fig. 6a). In contrast, the addition of NLS at the C-terminus of MCP-*Dr*BphP severely impaired its functionality, possibly due to interference with *Dr*BphP-LDB3 binding (Supplementary Fig. 6b). Moreover, targeting *HBG* with two sgRNAs and adjusting the expression levels of each module further increased *HBG* activation to 1577-fold under red light conditions (Supplementary Fig. 6c, d).

To evaluate OptoHEAL performance, we measured *HBG* mRNA expression in transfected HEK-293T cells under varying intensities (0-2 mW/cm^2) and duration (0-24 h) of red light (660 nm) illumination. QPCR-based analysis indicated that OptoHEAL exhibited both illumination time- and intensity-dependent activation of *HBG* expression (Fig. 4d, e). Notably, the OptoHEAL system showed negligible

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background leakage under dark conditions, while brief illumination (as short as 0.01 h) was sufficient to trigger significant gene activation (Fig. 4d). To test whether the OptoHEAL system could be switched on and off by alternating red/far-red light illumination, transfected HEK-293T cells were exposed to red light alone or red light followed by far-red light (Fig. 4f). We found that transcriptional activation was indeed reversible with this alternating red/far-red exposure regimen (Fig. 4g), thus providing a switchable ON-OFF control for manipulating gene expression. Moreover, OptoHEAL could activate other endogenous genes (*i.e.*, *UCP1*, *ASCL1*, and *IFNγ*) in HEK-293T cells (Fig. 4h). To explore its versatility for application in varying contexts, we transfected the OptoHEAL system into four additional mammalian cell lines and found that it could effectively activate *HBG* in all tested cell lines (Fig. 4i). These findings thus demonstrated OptoHEAL functions as a

Fig. 2 | Benchmarking of HEAL performance against state-of-the-art CRISPRa systems in mammalian cells and mice. a Schematics for endogenous gene activation with the HEAL (top), dCasMINI (middle), or dCpf1 (bottom) systems. In the HEAL system, transactivators are recruited to the MS2 aptamers within the sgRNA, whereas transactivators are fused directly to the CRISPR-dCas proteins in the dCasMINI and dCpf1 systems. **b**–**f** Evaluation of endogenous gene activation by HEAL in comparison with dCasMINI and dCpf1 systems in mammalian cells. B16F10 or HEK-293T cells were co-transfected with plasmids encoding HEAL, dCasMINI, or dCpf1 systems targeting different endogenous genes. QPCR assays of *Ucp1* (**b**) and *Tslp* (**c**) mRNA levels in B16F10 cells, as well as *RHOXF2* (**d**), *UCP1* (**e**), and *HBG* (**f**) transcription in HEK-293T cells were performed at 48 h post-transfection. In panels (**b**–**f**) colors indicate different systems: light blue, HEAL; orange-brown, dCasMINI; charcoal brown, dCpf1. **g** Schematic for endogenous gene activation with the HEAL or dCas9-SAM systems. In both systems, transactivators are recruited to the MS2 aptamers within the sgRNA. **h**–**k** Comparison of endogenous gene activation levels between the HEAL and dCas9-SAM systems. B16F10 or HEK-293T cells were co-transfected with plasmids encoding HEAL or dCas9-SAM systems targeting different endogenous genes. QPCR-based measurements of *Tslp* (**h**) and *Ucp1* (**i**) transcription in B16F10 cells, as well as *HBG* (**g**) and *UCP1* (**k**) in HEK-293T cells at 48 h post-transfection. In panels (**h**–**k**), colors indicate different systems: light blue, HEAL; beige, dCas9-SAM. **l** Schematic of the timeline for assessing endogenous gene activation in mice. Plasmids encoding the dCpf1, dCasMINI, HEAL, or dCas9-SAM system were delivered to mouse livers via hydrodynamic tail vein injection. NT, non-targeting. **m** Relative *Ucp1* mRNA expression in mouse livers following activation by the dCpf1, dCasMINI, HEAL, or dCas9-SAM systems. Data in (**b**–**f**, **h**–**k**) are means ± s.d.; $n = 3$ independent experiments, with 3 technical replicates for each. Data in (**m**) are means ± s.e.m.; $n = 6$ mice. Statistical significance was determined by two-way ANOVA (**b**–**f**, **m**) or one-way ANOVA (**h**–**k**); ns, not significant, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

red-light-inducible transcriptional activation tool with broad cellular compatibility.

To assess whether OptoHEAL could be applied for red-light-inducible transcriptional activation in vivo, we evaluated its ability to target our exogenous reporter system following delivery by hydrodynamic tail vein injection in C57BL6 mice. At ten hours post-injection, mice were illuminated with red light (660 nm, 20 mW/cm^2) for durations ranging from 0 to 10 h, then incubated in darkness for the remaining experimental period (Supplementary Fig. 7a). Bioluminescence imaging demonstrated light-dependent reporter activation, with luciferase expression increasing by up to 127-fold after 10 h of illumination (Supplementary Fig. 7b, c). Moreover, we used OptoHEAL to target the endogenous mouse gene, *Oct4*. Mouse livers were harvested at 24 h post-injection. Subsequent qPCR assays revealed that *Oct4* expression was indeed significantly upregulated (by 11.1-fold) following 10 h under red light illumination (Fig. 4j, k). Collectively, these results demonstrated that OptoHEAL functions as designed, enabling red-light inducible, adjustable, and reversible activation of endogenous genes both in vitro and in vivo.

Chemically inducible activation of endogenous genes

While the above data showed that the OptoHEAL system enables precise and reversible control of gene activation through light stimulation, its application may be constrained in physiological or clinical contexts where light delivery is infeasible or limited by poor tissue penetration. To address these challenges and expand its in vivo applicability, we developed a chemically inducible HEAL system (ChemHEAL) that allows controlled activation of endogenous genes through exposure to small molecules. Similar to the design principle for OptoHEAL, ChemHEAL was engineered by fusing MCP and p65-HSF1 modules with small molecule-responsive dimerization proteins. That is, in the presence of a specific small molecule, the activator, p65-HSF1, is recruited to the target promoter to initiate transcription of endogenous genes (Fig. 5a–e).

To demonstrate the adaptability of the ChemHEAL platform to various chemical agents and biological contexts, we constructed multiple ChemHEAL variants that could respond to different small molecules, including the plant hormone abscisic acid (ABA) and FDA-approved drugs such as grazoprevir (GZV), danoprevir (DNV), and rapamycin (Rapa). By integrating the transcriptional activation modules (MCP and p65-HSF1) with small molecule-responsive protein dimerization modules (ABI/PYL1 for ABA, GNCR1/NS3a for GZV, DNCR2/NS3a for DNV, and FRB/FKBP for Rapa), we identified the configurations that could induce the highest activation of *HBG* in HEK-293T cells upon stimulation with the respective small molecules, with fold increases of 5501 (ABA), 4610 (GZV), 3104 (DNV), and 4354 (Rapa), respectively (Supplementary Fig. 8).

We then assessed ChemHEAL performance in targeting *HBG* following transfection of the respective ChemHEAL constructs into HEK-293T cells. At 24 h post-transfection, cells were incubated with different concentrations of ABA (Fig. 5f), GZV (Fig. 5g), DNV (Fig. 5h), or Rapa (Fig. 5i) for 24 h. QPCR analysis demonstrated dose-dependent activation of *HBG* transcription upon exposure to each small molecule. We also monitored *HBG* mRNA levels at different time points during small molecule treatment. Across all four ChemHEAL systems, mRNA levels of *HBG* progressively increased over 48 h (Fig. 5j–m). Notably, mRNA expression significantly increased within 2 h of induction, highlighting the rapid responsiveness of the ChemHEAL platform. Tests of ChemHEAL in various mammalian cell lines demonstrated that it could efficiently activate *HBG* in all examined cell types, with induction levels ranging from 60- to 3584-fold (Supplementary Fig. 9). By delivering sgRNAs targeting *UCP1*, *ASCL1*, or *IFNγ*, we observed that all four ChemHEAL systems could activate each respective gene in HEK-293T cells, thus demonstrating its broad applicability across different gene targets (Fig. 5n–q). We next benchmarked the inducible HEAL variants against the constitutive system. Both OptoHEAL and ChemHEAL mediated stimulus-dependent transcriptional activation in response to light or small molecules, with moderately reduced maximal efficiencies compared to the constitutively active HEAL (Supplementary Fig. 10). This attenuated yet tightly regulated activity represents a deliberate design optimization to enhance safety and controllability.

To evaluate the chemical inducibility of transcriptional activation by ChemHEAL in vivo, we used the ABA inducible ChemHEAL system targeting our luciferase reporter as a representative model. Plasmids encoding ChemHEAL with the luciferase reporter were delivered to mouse livers by hydrodynamic injection via the tail vein. At 10 h post-injection, mice received intraperitoneal ABA (0-100 mg/kg) injection (Supplementary Fig. 11a). Bioluminescence imaging at 14 h post-induction revealed a dose-dependent increase in luciferase signal, reaching up to a 40-fold enhancement at 100 mg/kg compared to the 0 mg/kg control (Supplementary Fig. 11b, c). Taken together, these results showed that ChemHEAL could be applied in vitro or in vivo for rapid, sensitive, small molecule-inducible gene activation.

ChemHEAL-mediated *Tslp* activation can reduce body weight in HFD-induced obese mice

To explore the therapeutic potential of ChemHEAL, we employed the ABA-inducible ChemHEAL system to regulate *Tslp* expression, a cytokine well-known to alleviate obesity and related complications[49]. However, excessive *Tslp* expression can result in airway inflammation and hyperreactivity[50,51]. To achieve precise and controllable activation of *Tslp*, the *Tslp*-targeting ChemHEAL system was packaged into two hepatotropic AAV-2/8 vectors (AAV_{Tslp}) and delivered to the livers of high-fat diet (HFD)-induced obese mice via tail vein injection. Control mice were administered AAV_{NT} targeting an exogenous luciferase reporter instead of *Tslp*. At two weeks post-injection, AAV-treated obese mice received intraperitoneal injections of ABA (100 mg/kg) every two days for eight weeks (G3: AAV_{NT} + ABA, G5: AAV_{Tslp} + ABA),

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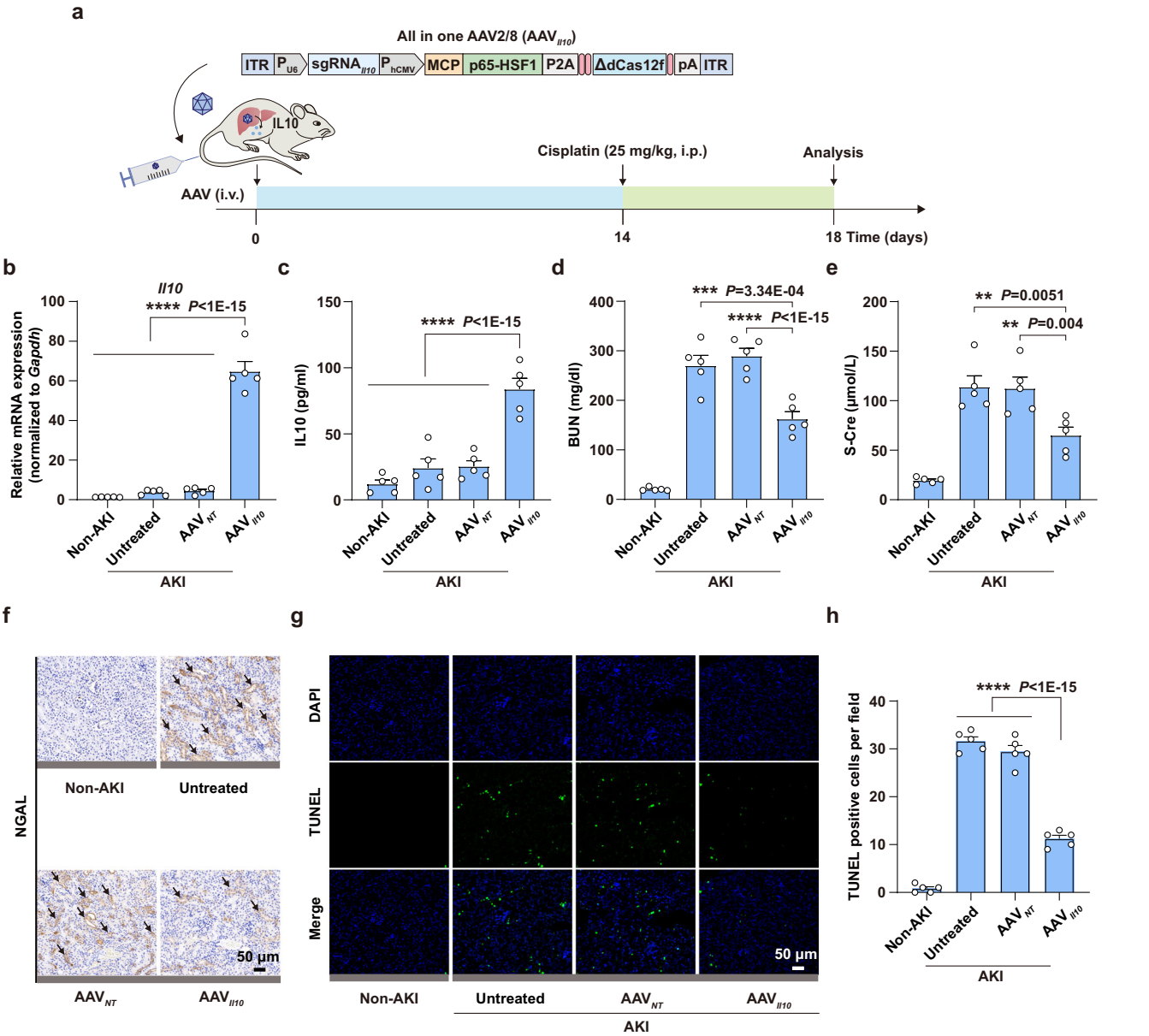


Fig. 3 | Transcriptional activation of endogenous *Il10* by HEAL ameliorates acute kidney injury in mice. a Schematic for activation of endogenous *Il10* using HEAL in acute kidney injury (AKI) mice. C57BL6 mice were administered with an AAV encoding the HEAL system (3×10^{11} vg) via intravenous injection (i.v.). After two weeks, AKI was induced by a single intraperitoneal (i.p.) injection of cisplatin (25 mg/kg). Four days after cisplatin administration, renal injury markers were analyzed, including serum blood urea nitrogen (BUN), serum creatinine (S-Cre), and neutrophil gelatinase-associated lipocalin (NGAL) in renal tissue. Non-AKI mice (non-AKI), untreated AKI mice (untreated), and AKI mice treated with AAV$_{NT}$ (AAV-delivered HEAL system with an empty sgRNA vector) served as controls. **b** Relative mRNA expression of *Il10* in mouse livers. **c** Serum IL-10 levels. **d**, **e** Serum levels of BUN (**d**) and S-Cre (**e**). **f** Representative images of immunohistochemistry (IHC) staining for neutrophil gelatinase associated lipocalin (NGAL) in renal tissue. The images represent typical results from three independent measurements. Black arrows, NGAL-positive regions. Scale bars = 50 μm. **g**, **h** TUNEL staining (**g**) and quantitative analysis of apoptotic cells (**h**) in renal tissue. The images represent typical results from three independent measurements. Cell nuclei were stained with DAPI. Scale bars = 50 μm, field = 0.25 mm^2. Data in (**b**–**e**, **h**) are means ± s.e.m.; $n = 5$ mice. Statistical significance was determined by one-way ANOVA (**b**–**e**, **h**); $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

or PBS (G4: AAV$_{Tslp}$ - ABA), while non-obese mice (G1) and untreated obese mice (G2) served as additional control groups (Fig. 6a).

At eight weeks post-ABA injection, *ΔdCas12f* mRNA could be detected in liver tissues of all AAV-injected groups, thus validating the success of viral transduction (Fig. 6b). In contrast, assessment of *Tslp*/TSLP at the mRNA and protein levels demonstrated that only ABA-treated AAV$_{Tslp}$ mice showed significant increases in *Tslp*/TSLP expression (Fig. 6c, d), indicating that ChemHEAL effectively activated *Tslp* in a small molecule-specific manner. Within two weeks of ABA treatment, mice in the AAV$_{Tslp}$ + ABA group exhibited a significant reduction in body weight compared to the AAV$_{NT}$ + ABA, AAV$_{Tslp}$ - ABA, and untreated obese groups. By eight weeks post-administration, the body weight of mice in the AAV$_{Tslp}$ + ABA group had decreased to an average of 36.1 g, whereas untreated obese mice exhibited a continued increase in body weight, reaching an average of 51.9 g (Fig. 6e, f). In addition, mice in the AAV$_{Tslp}$ + ABA group had significantly reduced adipose tissue mass compared to untreated obese controls, with epididymal white adipose tissue (eWAT) decreasing from 2.86 ± 0.30 g to 0.92 ± 0.33 g, subcutaneous inguinal white adipose tissue (iWAT) from 2.72 ± 0.15 g to 0.73 ± 0.47 g, and brown adipose tissue (BAT) from

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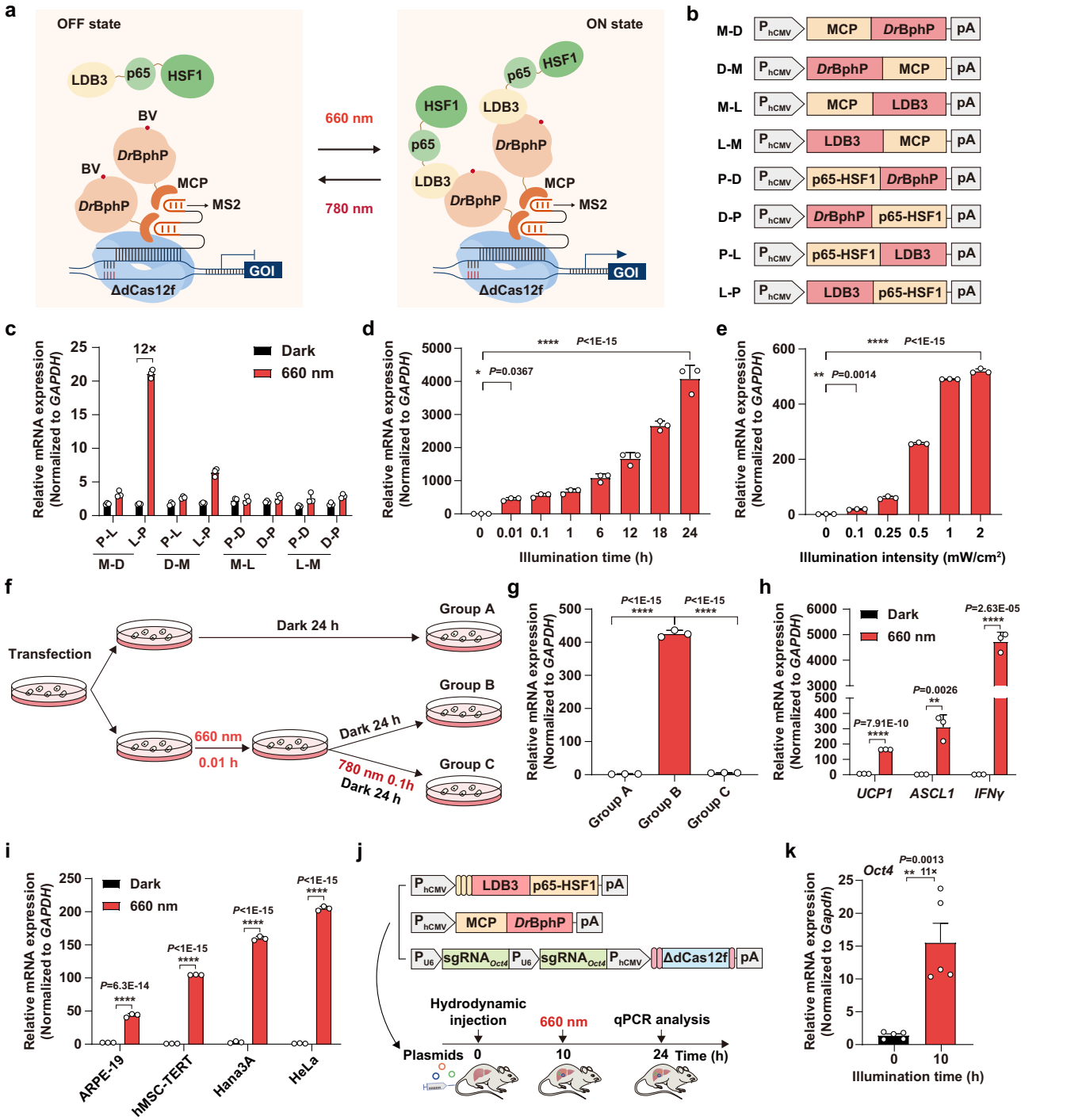


Fig. 4 | A red light-inducible HEAL system (OptoHEAL) for optogenetic regulation of endogenous gene transcription. a Schematic of the OptoHEAL system. In dark conditions, the ΔdCas12f-sgRNA complex targets the gene of interest (GOI), while *Dr*Bphp, fused with MCP, binds to the MS2 aptamers. Upon red light (660 nm) exposure, the nanobody (LDB3), fused to transactivators (p65-HSF1), binds specifically to *Dr*Bphp-MCP, activating endogenous gene transcription. Illumination with far-red light (780 nm) induces LDB3 dissociation from *Dr*Bphp, halting GOI transcription. **b** Schematic for different configurations of the photo-responsive components. One-letter abbreviations at left indicate genetic configurations. **c** Relative *HBG* mRNA levels induced by OptoHEAL variants. **d** Illumination time-dependent activation of *HBG* mediated by OptoHEAL system. **e** Illumination intensity-dependent activation of *HBG* mediated by OptoHEAL system. **f** Schematic of the timeline for ON/OFF switch performance of the OptoHEAL system. HEK-293T cells transfected with OptoHEAL targeting *HBG* were divided into three groups. Group A was kept in constant darkness. Groups B and C were exposed to red light (660 nm, 2 mW/cm^2) for 0.01 h to switch on the system. Group B was then kept in darkness for 24 h, while group C was exposed to far-red light (780 nm, 2 mW/cm^2) for 0.1 h to switch off the system before returned to darkness for 24 h. **g** QPCR analysis of OptoHEAL-mediated *HBG* transcription as described in (**f**). **h** OptoHEAL-mediated activation of different endogenous genes in HEK-293T cells. **i** OptoHEAL-mediated *HBG* activation in diverse mammalian cell lines. **j** Schematic of the timeline for OptoHEAL-mediated activation of endogenous gene in mice. **k** QPCR analysis of *Oct4* activation in mice mediated by the OptoHEAL system. Data in (**c**–**e**, **g**–**i**) are means ± s.d.; $n = 3$ independent experiments, with 3 technical replicates for each. Data in (**k**) are means ± s.e.m.; $n = 5$ mice. Statistical significance was determined by one-way ANOVA (**d**, **e**, **g**), two-tailed unpaired *t* test (**h**, **k**), or two-way ANOVA (**i**); ns, not significant, $^{**}P < 0.01$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

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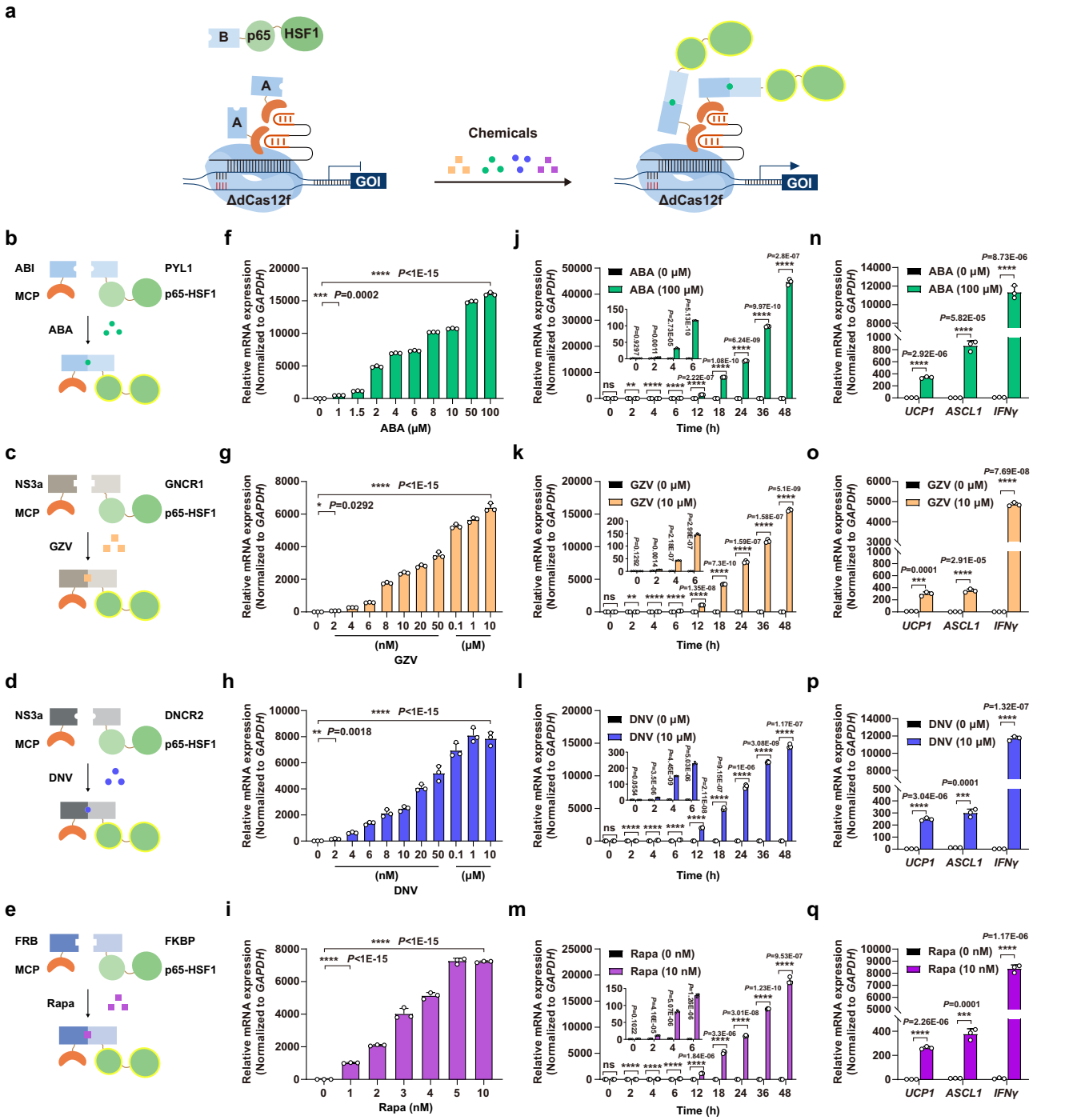
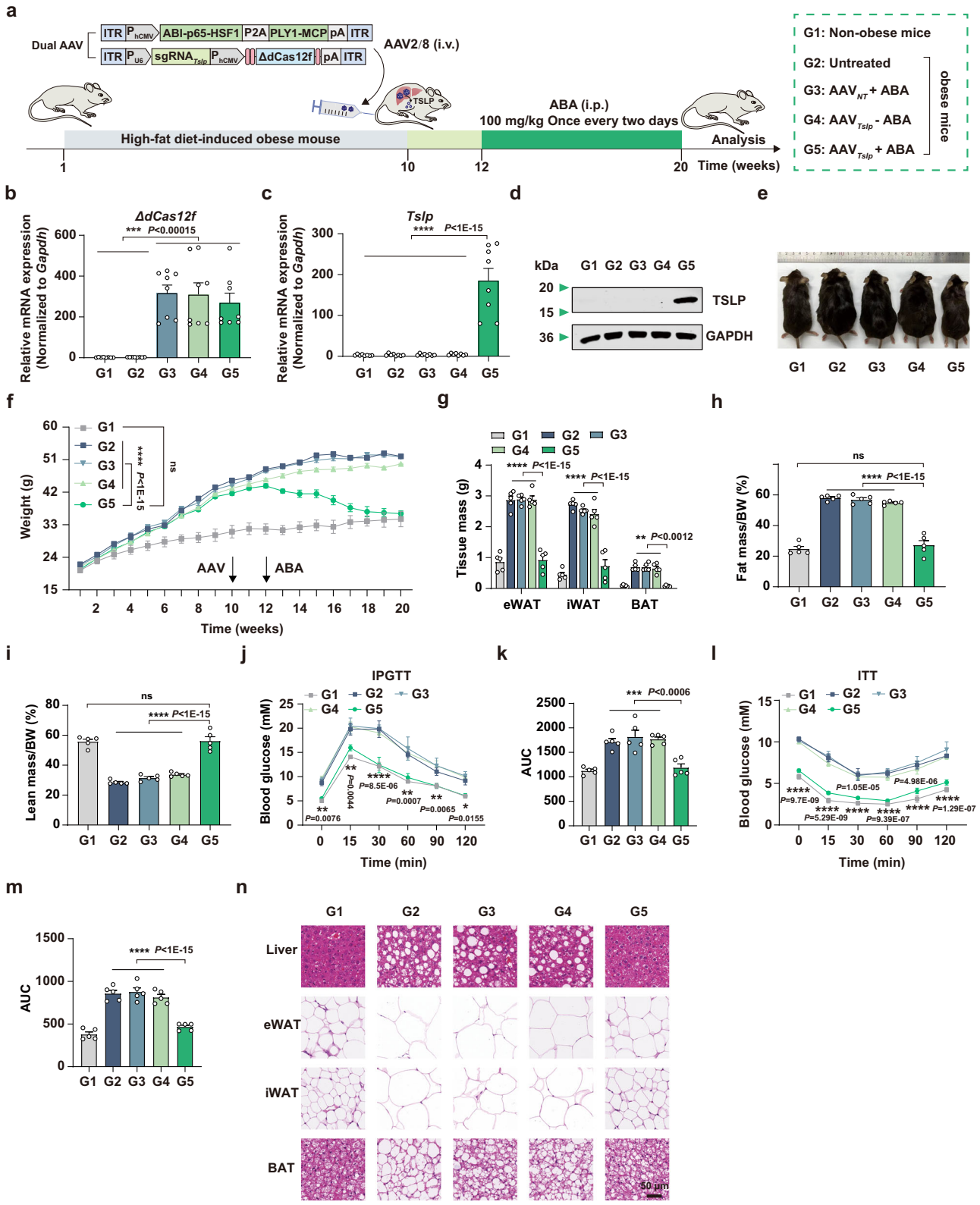


Fig. 5 | Chemical-inducible HEAL systems (ChemHEAL) for regulation of endogenous gene transcription. a Schematic of the ChemHEAL systems. In the presence of chemicals, specific fusion protein partners are recruited to each other, enabling the transactivator to localize to the gene of interest (GOI) and facilitate transcription of the endogenous gene with the assistance of the ΔdCas12f-sgRNA complex. **b**–**e** MCP and p65-HSF1 fragments were fused to chemical-inducible dimerization modules (ABI/PYL1 (**a**), NS3a/GNCR1 (**b**), NS3a/DNCR2 (**c**), or FRB/FRKB (**d**)). **f**–**i** Dose-dependent activation of *HBG* mediated by ChemHEAL system. HEK-293T cells transfected with plasmids encoding different ChemHEAL systems targeting *HBG* were treated with different concentrations of ABA (**f**), GZV (**g**), DNV (**h**), or Rapa (**i**) for 24 h. Relative mRNA levels of *HBG* were quantified using qPCR after induction. **j**–**m** Time-dependent activation of *HBG* mediated by ChemHEAL system. HEK-293T cells transfected with plasmids encoding different ChemHEAL systems targeting *HBG* were treated with 100 μM ABA (**j**), 10 μM GZV (**k**), 10 μM DNV (**l**), or 10 nM Rapa (**m**), and the relative mRNA levels of *HBG* were quantified by qPCR at different time points (0–48 h). **n**–**q** ChemHEAL-mediated activation of different endogenous genes. HEK-293T cells transfected with plasmids encoding different ChemHEAL systems targeting *UCP1*, *ASCL1*, or *IFNγ* were treated with 100 μM ABA (**n**), 10 μM GZV (**o**), 10 μM DNV (**p**), or 10 nM Rapa (**q**) for 24 h. Relative mRNA levels were quantified using qPCR after induction. Data in (**f**–**q**) are means ± s.d.; $n = 3$ independent experiments, with 3 technical replicates for each. Statistical significance was determined by two-tailed unpaired *t* test (**f**–**q**); ns, not significant, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

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0.70 ± 0.12 g to 0.08 ± 0.02 g (Fig. 6g). Moreover, the fat mass-to-body weight ratio was significantly lower in the AAV$_{Tslp}$ + ABA group, while lean mass was increased (Fig. 6h, i).

Metabolic parameters, including intraperitoneal glucose tolerance test (IPGTT), insulin tolerance test (ITT), and measurements of serum total cholesterol (TC), and triglycerides (TG), demonstrated that glucose metabolism and lipid profiles in the AAV$_{Tslp}$ + ABA group improved significantly, closely resembling those of non-obese mice by eight weeks after treatment (Fig. 6j–m and Supplementary Fig. 12a, b). Histological analysis by hematoxylin and eosin (H&E) staining revealed that adipocyte sizes in adipose tissues (eWAT, iWAT, and BAT) were significantly reduced in the AAV$_{Tslp}$ + ABA group. Moreover, ABA-treated AAV$_{Tslp}$ mice had notably lower severity of hepatic steatosis compared with control groups, suggesting improved liver health (Fig. 6n). Long-term treatment with AAV$_{Tslp}$ and ABA did not result in detectable adverse effects, as evidenced by normal lung histology,

Fig. 6 | Transcriptional activation of endogenous *Tslp* by ChemHEAL reduces body weight in high-fat diet (HFD)-induced obese mice. a Schematic of the experimental procedure and time schedule for AAV-delivered ChemHEAL-mediated *Tslp* activation in HFD-induced obese mice. C57BL6 mice were fed with HFD for 10 weeks to establish an obesity model, followed by intravenous injection of a mixture of dual-AAV encoding ABA-inducible ChemHEAL system (4×10^{11} vg were used for each virus) via the tail vein. Two weeks later, AAV_{Tslp}-treated obese mice were intraperitoneally injected with ABA (0 or 100 mg/kg) every two days for 8 weeks (G4: AAV_{Tslp} - ABA, G5: AAV_{Tslp} + ABA). Intraperitoneal glucose tolerance tests (IPGTT) and insulin tolerance tests (ITT) were performed before euthanizing the mice at 20 weeks for further analyses. Non-obese mice on a normal diet (G1), untreated obese mice (G2), and obese mice treated with AAV_{NT} (AAV-delivered ChemHEAL system with an empty sgRNA vector) followed by ABA treatment (G3: AAV_{NT} + ABA) served as controls. **b** Relative mRNA levels of *ΔdCas12f* in mouse liver. **c** Relative mRNA levels of *Tslp* in mouse liver. **d** Immunoblotting analysis of TSLP protein in mouse liver. Representative images from three independent experiments are shown. **e** Representative images of mice at 20 weeks. **f**–**m** Metabolic parameters assessments in mice treated as described in (**a**) including body weight (**f**), adipose tissue (eWAT, iWAT, and BAT) weights (**g**), ratio of fat and lean weights to body weight (**h**, **i**), IPGTT (**j**) and AUC analysis of the IPGTT data (**k**), ITT (**l**) and AUC analysis of the ITT data (**m**). **n** Representative H&E staining of liver and adipose tissues. The images represent typical results from three independent measurements. Scale bars = 50 μm. Data in (**b**–**m**) are means ± s.e.m.; $n = 8$ (**b**, **c**) or 5 (**f**–**m**) mice. Statistical significance was determined by one-way ANOVA (**b**, **c**, **g**–**i**, **k**, **m**) or two-way ANOVA (**f**, **j**, **l**). *P*-values in (**j**, **l**) were calculated by comparing the AAV_{Tslp} - ABA group with the AAV_{Tslp} + ABA group; ns, not significant, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. Source data are provided as a Source Data file.

stable levels of Th2-type pro-inflammatory cytokines (IL-4, IL-5, and IL-13) in bronchoalveolar lavage fluid, and unchanged systemic inflammatory markers (IL-6, TNF-α, and IgG) (Supplementary Fig. 13). Collectively, these results demonstrated that the AAV-mediated ChemHEAL system could be used for precise control of endogenous gene activation in vivo, and thus provided a promising therapeutic strategy for obesity and metabolic disorders.

Discussion

In this study, we developed a compact and efficient CRISPRa platform that couples an engineered, catalytically inactive Cas12f (ΔdCas12f) with an RNA aptamer-based transactivation strategy to enable potent activation of endogenous genes. HEAL could induce up to 100,000-fold increase in transcription of endogenous genes, substantially exceeding other current CRISPRa systems. In addition, its compact size allows full packaging within a single AAV vector, thereby overcoming a major barrier for in vivo delivery and therapeutic applications. To enable remote and precise control, we developed OptoHEAL and ChemHEAL, which are respectively inducible with red light and small molecules. Other existing controllable CRISPRa systems based on dCas9 or dCas12a typically face constraints on their delivery due to their large sizes. For instance, the photoactivatable $REDMAP_{Cas}$ system requires light-induced expression of MCP-p65-HSF1[36], and the red/far-red light-inducible $REDLIP_{Cas}$ system depends on dCas9-mediated recruitment[48], both of which increase system complexity and limit activation efficiency. In contrast, OptoHEAL enables rapid, reversible, and low-leakage gene activation while retaining a compact design. ChemHEAL incorporates modules that are specifically responsive to FDA-approved drugs (GZV, DNV, and Rapa) or a natural plant hormone (ABA), thus extending its applicability to light-inaccessible or systemically regulated contexts.

For genes with stringent expression thresholds or dosage-sensitive functions, excessive or sustained activation can result in cytotoxicity or developmental abnormalities. Therefore, the ability of OptoHEAL and ChemHEAL to provide precise, reversible, and tunable control over gene activation offers a significant advantage for therapeutic applications. This highlights the importance of regulatory precision in clinically oriented gene modulation systems. By extending the CRISPRa toolkit, OptoHEAL and ChemHEAL enable temporal and dose-dependent control of endogenous gene activation, potentially enhancing safety by preventing sustained or irreversible expression.

In the context of regenerative medicine, OptoHEAL could be applied for spatially restricted activation of neural transcription factors like NEUROG2 in induced pluripotent stem cells (iPSCs)[34], facilitating patterned activation within organoids and supporting temporally staged differentiation while minimizing off-target effects or premature lineage commitment. For cancer immunotherapy, ChemHEAL has the potential to enable systemically tunable activation of tumor-associated antigens or immunomodulatory factors using clinically approved small molecules[52]. These systems allow flexible dosing to balance therapeutic efficacy with toxicity, providing a means to transiently and on-demand boost antitumor immunity without the risk of chronic overstimulation. Moreover, the modular design supports the adaptability of this platform to a wide range of target genes and cellular contexts, thus providing a powerful and versatile tool for applications in gene therapy, disease modeling, and regenerative medicine.

Despite these advantages, several challenges remain. Currently, the overall sizes of the OptoHEAL and ChemHEAL variants exceed the packaging capacity for a single AAV vector, which may limit their efficiency and practicality for in vivo applications. Future iterations could benefit from incorporating smaller regulatory components, e.g., light-responsive Amg2 (*AM1-C0023g2*)[53], TetR (tetracycline-responsive repressor) systems[54], or heat-inducible HSP (heat shock protein) promoters[55], to accommodate vector size requirements and enhance delivery efficiency. In addition, although HEAL could activate endogenous genes in the liver and alleviate acute kidney injury in mice, further studies are warranted to develop tissue-specific delivery strategies and broaden its utility to other organs and systems. Notably, the ability of OptoHEAL and ChemHEAL to modulate multiple genes with precise temporal control may prove valuable in future applications involving complex disease networks. For instance, in oncology or immunotherapy, such systems could potentially enable coordinated regulation of immune modulators or signaling pathways, a direction that warrants further investigation.

In summary, HEAL and its inducible variants, OptoHEAL and ChemHEAL, constitute a compact, modular, and finely tunable platform for precise activation of endogenous genes. By addressing key challenges in controllability and vector compatibility, these systems lay a strong foundation for in vivo gene regulation and hold promise for broad application in therapeutic development and precision medicine.

Methods

Plasmid construction

Comprehensive design and construction details for all expression plasmids are provided in Supplementary Data 1. The detailed DNA sequence information for HEAL components is provided in Supplementary Data 2. All plasmids were constructed using a MultiS One Step Cloning kit (Vazyme, C113-01). All constructs were verified by Sanger sequencing (Shanghai Saiheng Biotechnology).

Cell culture and transfection

Human embryonic kidney cells (HEK-293T; ATCC, CRL-11268), HEK-293 derived Hana3A cells engineered for constitutive expression of RTP1, RTP2, REEP1, and $G_{\alpha o \lambda \phi}$[56], telomerase-immortalized human mesenchymal stem cells (hMSC-TERT; ATCC, SCRC-4000), human cervical adenocarcinoma cells (HeLa; ATCC, CCL-2), mouse neuroblastoma neuro-2a cells (N2a; ATCC, CCL131) cells, and adult retinal pigment epithelial cell line-19 (ARPE-19; ATCC, CRL-2302) were

cultured in Dulbecco's modified Eagle medium (DMEM; Gibco, C11995500BT) supplemented with 10% (vol/vol) fetal bovine serum (FBS; AusGeneX Industries, FBSSA500-S) and 1% (vol/vol) penicillin/streptomycin solution (Beyotime, ST488-1/ST488-2). Mouse B16F10 melanoma cancer cell line (B16F10; ATCC, CRL-6475) were cultured in Roswell Park Memorial Institute 1640 Medium (RPMI1640; Gibco, 11875093) with 10% (v/v) FBS and 1% (v/v) penicillin/streptomycin. All cell lines were cultured at 37 °C in a humidified atmosphere containing 5% CO_2 and were regularly tested for the absence of *Mycoplasma* and bacterial contamination.

Cells were transfected using an optimized polyethyleneimine (PEI)-based protocol. Briefly, 8×10^4 cells were seeded per well in a 24-well plate and cultured overnight to achieve 60–70% confluency. Subsequently, cells were incubated for 6 h with 50 µl of PEI (Polysciences, 4765; molecular weight 40,000, stock solution 1 mg/ml in ddH_2O) and corresponding plasmid mixture at a mass ratio of 3:1. For ARPE-19 and HEK-293T cells, transfections were performed at 70–80% confluency using Lipofectamine™ 8000 Transfection Reagent (Beyotime, C0533FT) according to the manufacturer's protocol. N2a cells were transfected at 70–80% confluency using Lipofectamine™ 3000 Transfection Reagent (Thermo Fisher Scientific, L3000075) according to the manufacturer's protocol. Detailed descriptions of transfection mixtures are provided in Supplementary Data 3 and Supplementary Data 4.

Quantitative PCR (qPCR) analysis

Cells and tissue samples were harvested for total RNA isolation using an RNAiso Plus Kit (Takara, 9108) according to the manufacturer's instructions. A total of 1 µg RNA was reverse-transcribed into complementary DNA using a HiScript® III RT SuperMix with the genomic DNA Eraser (Vazyme, R323). QPCR was performed using the ChamQ Universal SYBR qPCR Master Mix (Vazyme, Q711-02) on the LightCycler 96 Real-Time PCR Instrument (Roche). Primer sequences are listed in Supplementary Table 1, and the target sequences of sgRNAs used in this study are listed in Supplementary Data 5. Gene expression levels were normalized to human/mouse glyceraldehyde 3-phosphate dehydrogenase coding gene (*GAPDH*), and relative mRNA abundance was calculated using the $2^{-\Delta\Delta Ct}$ method.

Enzyme-linked immunosorbent assay (ELISA)

IL-10, TSLP, and IFNγ levels in cell culture supernatants were quantified using the mouse IL-10 ELISA kit (Invitrogen, 88-7105-88), the mouse TSLP ELISA kit (R&D, MTLP00), and the human IFNγ ELISA kit (MultiSciences Biotech, EK180-96), respectively. IL-4, IL-5, and IL-13 levels in bronchoalveolar lavage fluid were quantified using the mouse IL-4 ELISA kit (Absin, abs520003), the mouse IL-5 high sensitivity ELISA kit (MultiSciences Biotech, EK205HS), and the mouse IL-13 ELISA kit (Invitrogen, KMC2221), respectively. Serum concentrations of IL-10, IL-6, and TNF-α were quantified using the mouse IL-10 ELISA kit (Invitrogen, 88-7105-88), the mouse IL-6 ELISA kit (MultiSciences Biotech, EK206), and the mouse TNF-α ELISA kit (MultiSciences Biotech, EK282), respectively. Serum IgG levels were quantified using the mouse IgG ELISA kit (MultiSciences Biotech, EK271). All ELISA assays were conducted according to the manufacturer's instructions.

OptoHEAL-mediated activation of endogenous gene transcription in mammalian cells

To evaluate light-controlled endogenous gene activation, 8×10^4 cells were plated in a 24-well plate and co-transfected with plasmids encoding the OptoHEAL system targeting the corresponding gene [pQL236 (P_{hCMV}-3 × SV40-NLS-LDB3-p65-HSF1-pA), pDQ400 (P_{hCMV}-MCP-*Dr*BphP-pA), pWH299 (P_{hCMV}-2 × c-MYC NLS-dUn1Cas12f (D143R/T147R/K330R)−1 × c-MYC NLS-pA), and sgRNAs targeting the corresponding gene]. Twenty-four hours after transfection, the cell culture plate was placed below a custom-designed 4 × 6 red light LED array (660 nm; Shenzhen Bested Opto-electronic; each LED centered above a single well), and the cells were exposed to red light (660 nm, 2 mW/cm², 24 h, unless indicated specifically). The light intensity was measured at a wavelength of 660 nm using an optical power meter (Advantest, Q8230). Relative mRNA expression levels were quantified using qPCR analysis 24 h after illumination.

To evaluate the tunability of the OptoHEAL system, HEK-293T cells were co-transfected with plasmids encoding the OptoHEAL system targeting the *HBG* locus [pQL236 (P_{hCMV}-3 × SV40 NLS-LDB3-p65-HSF1-pA), pDQ400 (P_{hCMV}-MCP-*Dr*BphP-pA), pWH299 (P_{hCMV}-2 × c-MYC NLS-dUn1Cas12f (D143R/T147R/K330R)−1 × c-MYC NLS-pA), pWH187 (P_{U6}-sgRNA$_{HBG}$ (ES1 + 2 + 5)-pA), pWH190 (P_{U6}-sgRNA$_{HBG}$-2-pA)]. Twenty-four hours after transfection, cells were divided into three groups. Group A was maintained in darkness throughout the experiment (OFF), group B was exposed to red light (660 nm, 2 mW/cm²) for 0.01 h (ON), and group C was illuminated with red light (660 nm, 2 mW/cm²) for 0.01 h (ON), followed by far-red light (780 nm, 2 mW/cm²) for 0.1 h (OFF). Relative mRNA expression levels were quantified using qPCR analysis 24 h after initial light exposure.

ChemHEAL-mediated activation of endogenous gene transcription in mammalian cells

To evaluate chemically controlled endogenous gene activation, 8×10^4 cells were plated in a 24-well plate and co-transfected with plasmids encoding the ChemHEAL system targeting the corresponding gene [pWH299 (P_{hCMV}-2 × c-MYC NLS-dUn1Cas12f (D143R/T147R/K330R)−1 × c-MYC NLS-pA), pWH253 (P_{hCMV}-PYL1-MCP-pA), pWH255 (P_{hCMV}-ABI-p65-HSF1-pA), pWH260 (P_{hCMV}-GNCR1-MCP-pA), pWH261 (P_{hCMV}-NS3a-MCP-pA), pWH265 (P_{hCMV}-NS3a-p65-HSF1-pA), pDQ601 (P_{hCMV}-MCP-FKBP-pA), pDQ603 (P_{hCMV}-p65-HSF1-FRB-pA), pDQ610 (P_{hCMV}-DNCR2-p65-HSF1-pA), and sgRNAs targeting the corresponding gene]. Twenty-four hours after transfection, the culture medium was refreshed, and the cells were incubated with ABA (100 µM; Sigma-Aldrich, 90769), GZV (10 µM; Adooq Bioscience, A173303), DNV (10 µM; Adooq Bioscience, RG7227), or Rapa (10 nM; MCE, HY-10219). Relative mRNA expression levels were quantified using qPCR analysis after 24 h of induction.

Animals

All animals were approved by the Institutional Animal Care and Use Committee of Shanghai and conducted in accordance with the National Research Council Guide for the Care and Use of Laboratory Animals. Male C57BL/6 mice (4–6 weeks old) were obtained from the East China Normal University (ECNU) Laboratory Animal Center. Sex was not considered as a biological variable in this study; all experiments were conducted using age-matched male mice only to minimize hormonal variability and ensure consistency in obesity-related phenotypes. Mice were kept under a standard alternating 12 h light/12 h dark cycles at 23 °C ± 1 °C with 60% ± 5% relative humidity, with free access to water and food. Obese mice were fed a high-fat diet containing 60 kcal% fat (Research Diets, D12492), while non-obese mice received a standard chow diet with a composition of 15 kcal% fat (Xietong Biology, XTI01CR-010).

In vivo bioluminescence imaging

Each mouse was intraperitoneally injected with D-Luciferin solution (150 mg/kg; Shanghai Sciencelight Biology Science & Technology, luc001). Five minutes after Luciferin injection, bioluminescence images of the mice were captured using the IVIS Lumina II in vivo imaging system (Perkin Elmer). Regions of interest were quantified using the Living Image software (version 4.3.1).

Comparison of the HEAL system with other transcriptional activation systems in mice

For exogenous gene activation experiments, 6-week-old male C57BL/6 mice were hydrodynamically injected with 2 mL of Ringer's solution

(0.147 M NaCl, 1.13 M $CaCl_2$, 4×10^{-3} M KCl) containing plasmids encoding the corresponding transcriptional activation systems (HEAL, dCasMINI, dCpf1, or dCas9-SAM) targeting either a luciferase reporter or a non-targeting control site. Twenty-four hours after plasmid injection, bioluminescence imaging was performed.

For the endogenous gene activation experiment, 6-week-old male C57BL/6 mice were hydrodynamically injected with 2 mL of Ringer's solution containing plasmids encoding the corresponding transcriptional activation systems targeting either the *Ucp1* locus or a non-targeting control site. Forty-eight hours after plasmid injection, mice were euthanized, and liver tissues were collected for qPCR analysis of *Ucp1* expression.

OptoHEAL/ChemHEAL-mediated activation of gene transcription in mice

For exogenous gene activation experiments, 6-week-old male C57BL/6 mice were hydrodynamically injected with 2 ml of Ringer's solution containing plasmids encoding either the OptoHEAL or ChemHEAL system targeting an exogenous luciferase reporter. Ten hours after plasmid injection, mice receiving OptoHEAL were illuminated with red light (660 nm LED, 20 mW/cm^2) for different time periods (0 to 10 h), while mice receiving ChemHEAL were intraperitoneally injected with ABA at different doses (0 to 100 mg/kg). Twenty-four hours after plasmid injection, bioluminescence imaging was performed.

For the endogenous gene activation experiment, 6-week-old male C57BL/6 mice were hydrodynamically injected with 2 ml of Ringer's solution containing plasmids encoding the OptoHEAL system targeting the *Oct4* locus. Ten hours after plasmid injection, mice were illuminated with red light (660 nm LED, 20 mW/cm^2) for 10 h. Twenty-four hours after plasmid injection, mice were euthanized, and liver tissues were collected for qPCR analysis of *Oct4* expression.

AAV-delivered HEAL-mediated activation of *Il10* transcription in AKI mice

Recombinant AAV (serotype 2/8) was produced as previously described[57]. For in vivo delivery of the HEAL system, 6-week-old male C57BL/6 mice were intravenously injected with 3×10^{11} vg of either AAV-P_{U6}-sgRNA$_{Il10}$::P_{hCMV}-MCP-p65-HSF1-P2A-ΔdCas12f-pA (3×10^{11} vg) or a non-targeting control AAV-P_{U6}-sgRNA$_{NT}$::P_{hCMV}-MCP-p65-HSF1-P2A-ΔdCas12f-pA via the tail vein. Two weeks after AAV injection, mice were intraperitoneally injected with cisplatin to induce AKI. Briefly, mice were subjected to an 18 h period of fasting and water dehydration prior to induction and then intraperitoneally injected with cisplatin (MCE, HY-17394) at a dose of 25 mg/kg, after which food and water were restored. Healthy control mice underwent the same fasting and dehydration procedure but did not receive cisplatin injection. Four days post-cisplatin injection, mice were euthanized and, liver, kidney, and serum samples were collected. Serum samples were analyzed for IL-10 levels using the mouse IL-10 ELISA Kit (Invitrogen, 88-7105-88), and for blood urea nitrogen (BUN) and serum creatinine (S-Cre) levels using commercial assay kits (Changchun Huili Biotech, S03036 and S03076). Liver tissues were used to evaluate *Il10* mRNA expression by qPCR. Kidney tissues were used for immunohistochemical staining of NGAL and for terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay to assess apoptosis.

Immunohistochemical staining

Kidney tissues were fixed in 4% paraformaldehyde (Servicebio, G1101), embedded in paraffin, and cut into 5 μm thick slices. The sections were dewaxed and subjected to antigen retrieval under appropriate conditions. Endogenous peroxidase was blocked using 3% hydrogen peroxide for 25 min at room temperature, followed by washes with PBS. Non-specific binding was blocked with 3% bovine serum albumin (BSA) for 30 min at room temperature. Sections were incubated with rabbit anti-NGAL primary antibody (1:2000; Abcam, EPR21092) diluted in PBS overnight at 4 °C in a humidified chamber. After PBS washes, sections were incubated with Horseradish Peroxidase (HRP)-labeled secondary antibody (1:200; Servicebio, GB23303) for 50 min at room temperature. 3,3'-Diaminobenzidine (DAB) was used for color development, and the reaction was stopped by rinsing with tap water. Sections were then counterstained with hematoxylin.

TUNEL staining

Paraffin-embedded kidney tissue sections were treated with freshly prepared permeabilization solution (0.3% Triton X-100 in PBS) for 5 min at room temperature, and then incubated with fluorescein-labeled deoxyuridine triphosphate (green) for 60 min at 37 °C in a dark, humidified chamber. Fluorescence images were captured using a BX53 upright fluorescence microscope (DMI8; Leica). Apoptotic nuclei were visualized in green, while all nuclei were counterstained with DAPI and appeared blue.

AAV-delivered ChemHEAL-mediated activation of *Tslp* transcription in HFD-induced obese mice

Four-week-old male C57BL/6 mice were fed high-fat diets consisting of 60 kcal% fat (Research Diets, D12492) for 10 weeks until the body weight exceeded 40 g. Subsequently, HFD-induced obese mice were intravenously injected with a mixture of recombinant AAV (serotype 2/8) vectors containing AAV-P_{hCMV}-ABI-p65-HSF1-P2A-PYL1-MCP-pA (4×10^{11} vg) and either AAV-P_{U6}-sgRNA$_{Tslp}$::P_{hCMV}-ΔdCas12f-pA or AAV-P_{U6}-sgRNA$_{NT}$::P_{hCMV}-ΔdCas12f-pA (4×10^{11} vg) via the tail vein. Two weeks after AAV injection, mice were intraperitoneally injected with ABA (100 mg/kg) or PBS every two days for 8 weeks. Body weight was monitored weekly. At week 10 post-AAV administration, metabolic parameters, including intraperitoneal glucose tolerance test (IPGTT), insulin tolerance test (ITT), serum total cholesterol (TC) and triglycerides (TG), and body composition (ratio of fat and lean weights to body weight) were assessed. Body composition was measured using the AccuFat MRI system (MAG-MED, AccuFat-1050). Bronchoalveolar lavage fluid and serum were collected for analysis of Th2-type pro-inflammatory cytokines (IL-4, IL-5, and IL-13) and systemic inflammatory markers (IL-6, TNF-α, and IgG), respectively. Liver tissues were collected for analysis of *Tslp* mRNA expression (by qPCR) and TSLP protein levels (by immunoblotting). Hematoxylin and eosin (H&E) staining was performed on liver, lung, and adipose tissues to assess histopathological changes.

Bronchoalveolar lavage procedure

At week 8 of ABA treatment, mice were euthanized by carbon dioxide asphyxiation, and a small incision of the trachea was made to insert a 24-G lavage catheter. The catheter and trachea were secured with a cotton thread, and 0.5 mL of sterile PBS was instilled into the lungs three times using a 1 mL syringe. Bronchoalveolar lavage fluid was pooled and centrifuged at 300 × *g* for 5 min at 4 °C to pellet cells. The supernatant was collected and stored at − 80 °C for subsequent cytokine analysis.

Intraperitoneal glucose tolerance test (IPGTT) in mice

Mice were fasted overnight for 16 h, and intraperitoneally injected with aqueous 1.5 g/kg D-glucose. Blood glucose levels were measured from tail vein blood samples at 0, 15, 30, 60, 90, and 120 min after injection using a handheld glucometer (Exactive Easy III, MicroTech Medical). The 0 min sample was used to determine fasting plasma glucose levels. The trapezoidal rule determined the area under the curve (AUC) for IPGTT.

Insulin tolerance test (ITT) in mice

Mice were fasted for 4 h and intraperitoneally injected with insulin solution (Sigma, 11061-68-0) at a dose of 0.75 U/kg body weight. Blood

samples were obtained via the tail vein at 0, 15, 30, 60, 90, and 120 min after insulin injection.

Total cholesterol and triglycerides measurement

To measure total cholesterol (TC) and triglycerides (TG) in serum, mouse blood was obtained from the orbital venous plexus and centrifuged at 1000 × *g* for 10 min at 4 °C to isolate serum. The serum levels of TC and TG were then determined using a total cholesterol assay kit (Nanjing Jian Cheng Bioengineering Institute, A111-1-1) and a triglyceride assay kit (Nanjing Jian Cheng Bioengineering Institute, A110-1-1), respectively, according to the manufacturer's instructions.

Immunoblotting analysis

Mice were euthanized, and liver tissues were collected and homogenized using magnetic beads in RIPA lysis buffer (Yeasen, 20101ES60) supplemented with protease inhibitor cocktail (Beyotime, P1005) and phosphatase inhibitor mixture (Beyotime, P1045). Homogenates were centrifuged to collect the supernatants containing total protein. Proteins were separated by SDS-PAGE and then electro-transferred onto polyvinylidene difluoride (PVDF; Millipore, ISEQ00010) membranes. Membranes were blocked with 5% non-fat milk and incubated overnight at 4 °C with primary antibodies: rabbit polyclonal anti-TSLP (1:2000; Proteintech, 13778-1-AP) and rabbit monoclonal anti-GAPDH (1:3000; Yeasen, 30202ES40). After three washed with Tris-buffered saline with 0.05% tween-20 (Sangon, 9005-64-5), membranes were incubated with Alexa fluor-based goat anti-rabbit IgG (H + L) secondary antibody (1:25,000; Yeasen, 33119ES60) at room temperature for 1 h. Images were scanned using the Odyssey CLX imaging system (Licor).

Hematoxylin and eosin (H&E) staining of liver, lung, and adipose tissues

Liver, lung, and adipose tissues were fixed in 4% paraformaldehyde (Servicebio, G1101) overnight at room temperature. The fixed samples were sequentially dehydrated through graded ethanol solutions, cleaned in xylene, embedded in paraffin, and sliced into 4 μm-thick sections with a rotary microtome (Leica RM2235, Manual Rotary Microtome). H&E staining was performed using an H&E Staining Kit (Servicebio, G1005) according to the manufacturer's instructions. Stained sections were imaged using an upright microscope (BX53, Olympus).

Statistics & reproducibility

All in vitro data are presented as the mean ± s.d. of three independent biological replicates unless otherwise stated. For animal experiments, each treatment group consisted of randomly selected mice (n = 5-10), and the results are expressed as mean ± s.e.m. Sample sizes were not predetermined by statistical methods but were based on prior studies reporting similar experimental designs. Investigators were blinded to allocation during experiments and outcome assessment. No data were excluded from the analyses. Comparisons between groups two were performed using two-tailed unpaired t tests. One-way or two-way ANOVA was used to evaluate differences among multiple groups. Statistical analyses were performed using GraphPad (Prism v.8), and statistical significance was defined as $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, and $^{****}P < 0.0001$. n and P values are described in the figures or figure legends.

Ethical statement

The experiments involving animals were approved by the East China Normal University (ECNU) Animal Care and Use Committee and in direct accordance with the Ministry of Science and Technology of the People's Republic of China on Animal Care guidelines. The protocol (protocol ID: m20220505 and m20241205) was approved by the ECNU Animal Care and Use Committee. All animals were euthanized after the termination of the experiments.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data associated with this study are presented in the paper or the Supplementary Information. The data generated in this study are provided in the Source Data file. Sequences for plasmids used in the study are available in the Supplementary Data file. All genetic components related to this paper are available with a material transfer agreement and can be requested from H.Y. (hfye@bio.ecnu.edu.cn). Source data are provided in this paper.

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Author contributions

H.Y. conceived the project. H.Y., D.K., and H.W. designed the project. H.W., D.K., T.Y., Y.Z., M.L., X.M., T.Z., W.Z., X.L., and J.Y. performed the experimental work. H.Y., H.W., D.K., and N.G. analyzed the results and wrote the manuscript. All authors edited and approved the manuscript.

Competing interests

H.Y., H.W., and N.G. are inventors on patent applications (Chinese patent application number 2025118539236) submitted by ECNU. The remaining authors declare no competing interests.

Additional information

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